

Broker API comparison — documentation vs. real behaviour

19 September 2026 · research only — no code was changed.

Covers every broker Ayush sent, plus the two we already know first-hand: Lakshmishree (live since 15 Sep) and Fyers (compared on 18 Sep).

Three kinds of evidence are kept apart on purpose:

Marking	Meaning
Measured	Numbers taken on our own machines today, or during our live burn-in
Documented	Stated in the broker's own API documentation
Not stated	The documentation is silent — never guessed here

1. What we actually need

The engine is not a high-frequency system. It evaluates **once per minute** and places at most a few orders a day. So the ranking below weighs **cost**, **reliable daily login** and **honest order feedback** far above raw speed.

#	Requirement	Why
1	NSE and BSE options (NIFTY + SENSEX weeklies)	Thursday trades SENSEX
2	Market order, carry-forward product (NRML/MARGIN)	overnight carry is on
3	Order id back immediately + a way to confirm the fill	the engine sizes and books from the fill price
4	Positions + funds read, for reconcile and sizing	we compare broker vs engine every minute
5	Unattended daily login	nobody is at a keyboard at 09:10
6	Low, predictable brokerage	we trade small premiums; costs dominate
7	~10 API calls/minute is plenty	every broker below clears this easily
8	Market data not required from the broker	we run on TrueData

2. Latency — measured today, all brokers, same method

Unauthenticated requests from **our backend container** (the machine that sends the live orders), 19 Sep 2026 07:19 IST. 8 samples warm (keep-alive, which is what our adapter uses), 8 cold, 6 TLS handshakes. Nothing was ordered; no credentials used.

Broker	Warm median	Warm best	Cold median	TLS	Host IP	Note
Shoonya (Finvasia)	23.7 ms	15.3 ms	95.5 ms	36.3 ms	13.202.119.185	AWS Mumbai
Choice FinX	24.5 ms	18.5 ms	82.4 ms	46.0 ms	13.203.137.96	AWS Mumbai
Lakshmishree (ours)	29.5 ms	14.6 ms	61.9 ms	29.5 ms	202.149.222.131	own server, best cold time
Alice Blue	29.6 ms	27.8 ms	143.2 ms	36.5 ms	104.18.17.227	Cloudflare
Zerodha Kite	36.3 ms	31.0 ms	88.7 ms	39.7 ms	104.16.34.50	Cloudflare
Fyers v3	42.3 ms	40.1 ms	111.3 ms	32.8 ms	104.18.5.135	Cloudflare
Groww	45.1 ms	40.9 ms	180.0 ms	35.8 ms	104.18.37.214	Cloudflare
Upstox (reference)	45.2 ms	36.0 ms	164.6 ms	41.6 ms	172.64.147.179	Cloudflare
Dhan	52.9 ms	40.1 ms	135.7 ms	92.4 ms	108.158.46.79	AWS CloudFront
5paisa	70.1 ms	48.2 ms	155.1 ms	40.0 ms	23.55.244.130	Akamai
Angel One SmartAPI	71.2 ms	22.9 ms	93.0 ms	41.4 ms	103.82.178.77	own range, jittery
Sharekhan	101.2 ms	86.5 ms	403.1 ms	170.5 ms	159.60.135.114	slowest by far
Kotak Neo	—	—	—	—	does not resolve	see §4

How to read this. Anything inside ~20 ms of another is noise — a repeat run put Lakshmishree at 18.5 ms and Fyers at 56 ms. What is *not* noise: Sharekhan is 3–4× slower than the field on every metric, and Kotak's documented gateway hostname (`gw-napi.kotaksecurities.com`) **does not exist in DNS** at all.

None of this measures order matching at the exchange — that is the same exchange for everyone. It measures the pipe our adapter talks through.

3. Lakshmishree (Symphony XTS) — documentation vs. what we found

This is the section Ayush asked for specifically. Left column = the published XTS spec. Right column = what our own system did, with the evidence.

Topic	XTS documentation says	What we measured / hit in production
Error signalling	Errors return HTTP 400 / 404 / 429 / 500 with <code>{"type":"error",...}</code>	Mismatch — A dead token returns HTTP 200 with <code>{"type":"error","description":"Invalid Token"}</code> . Never 401/403. Anything keying on status code treats a dead session as success. We rewrote <code>_auth_rejected()</code> to match on the body; 16/16 regression tests
Session model	"Only one session is allowed per appKey–secretKey" — documented	Matches — true, and harsher than it reads: a second login silently kills the first. Recovery needed our own re-login-on-body-error path (~230–410 ms)
Token life	24 hours	Consistent with what we see; we re-login on rejection rather than on a timer
Login latency	not stated	Measured 65–81 ms
Funds / balance	<code>BalanceList[0].limitObject...</code>	Matches — shape correct — but Adhoc margin fields can be the literal string "NaN" , undocumented. Balance read ~71 ms median
Order placement	POST <code>/interactive/orders</code> , returns <code>AppOrderID</code> synchronously	Matches — correct. Measured place → fill confirmed: 1,908 ms (BUY) and 315 ms (SELL)
Fill confirmation	socket.io pushes <code>order / trade / position</code> events	Caveat — available but we don't use it — we poll order status 6× at 0.5 s. That polling, not the network, is most of the 1,908 ms
Rate limits	Orders 10/s shared; balance, profile, order book, trade book 1/s each ; 429 on breach	Never hit — we use ~1 order/minute. The 1/s limits on reads are tighter than most rivals, and worth knowing before adding dashboards
Positions / reconcile	<code>/interactive/portfolio/positions</code>	Matches — works. Measured 247 ms ; broker held 845 qty = engine's 845
Broker MTM / margin	fields exist	Mismatch — MTM and RealizedMTM stayed 0.00 , and margin utilised lagged minutes after the fill (showed ₹0 used right after a ₹971.75 buy). We now compute and display our own figures next to theirs
Cold start	not stated	First account refresh timed out at 13.7 s before the session was warm; now three reads run in parallel with one retry — refresh 499 ms / fetched in 261 ms
Instrument identity	<code>exchangeInstrumentID</code> , master via POST <code>/apimarketdata/instruments/master</code>	Caveat — That master needs an authenticated market-data session, which we do not have (we run TrueData). We fall back to Zerodha's public instrument dump , whose <code>exchange_token</code> equals the XTS id — an undocumented dependency we verified contract-by-contract
Market-data product	separate XTS MD API	Caveat — Its ~50-instrument cap wedged subscription swaps for us — one reason we moved the feed to TrueData
Brokerage	nothing in the API	Mismatch — The worst gap. The API reports no charges at all. The contract note showed ₹40 per lot per side , hidden inside the trade rates (bought 1.7654 vs our fill 1.15). Our cost model assumed ₹17/order → it under-counted a real trade 30× (₹210 modelled vs ₹1,397.95 actually charged)
Latency claims	none published	our own numbers above are the only figures that exist

Verdict on Lakshmishree: the API itself is **fast and correct** — the fastest cold connection of the whole field, order round trips under 2 s, reconcile that matched to the share. Its problems are **commercial and observational**, not technical: brokerage 30× what we modelled and invisible to the API, no cost or MTM reporting, and one undocumented behaviour (HTTP 200 errors) that could have silently disabled live trading.

4. The brokers Ayush sent — what the documentation says

Everything in this section is **documented**, not measured, unless marked.

4.1 Zerodha Kite Connect 3

- REST `api.kite.trade`, one socket `wss://ws.kite.trade` carrying **both** market data and **order updates**.
- Token dies **06:00 next day**; account must have TOTP 2FA; Zerodha's own position is that the daily login is **manual** — automating it is not permitted. Everything after that login is automatable.
- Orders: `POST /orders/:variety`, order id immediate; "placement does not imply execution".
- Limits: 10 orders/s, **400/min, 5,000/day**, 25 modifications per order; 3 sockets, 3,000 instruments each.
- Instruments: daily gzipped CSV; docs warn **tokens are reused after expiry** — key on `exchange:tradingsymbol`.
- **Cost: ₹500/month** (data included). Brokerage **₹20/order**.
- Static IP: not required.

4.2 DhanHQ v2

- REST `api.dhan.co/v2`, **separate order-update socket** `wss://api-order-update.dhan.co`, feed `wss://api-feed.dhan.co`.
- Token **24 h**, and — unusually — a **documented programmatic re-auth** when TOTP is enabled. API key/secret valid 12 months.
- Limits: orders 10/s, 250/min, 1,000/hr, **7,000/day**; feed 5,000 instruments/connection, 5 connections.
- **Static IP whitelisting is mandatory for place/modify/cancel**.
- **Cost: trading APIs free; market data ₹499/month**. Brokerage **₹20/order**.

4.3 Angel One SmartAPI

- REST `apiconnect.angelone.in`; feed `wss://smartapisocket.angelone.in`; **order updates** `wss://tns.angelone.in/smart-order-update`.
- Login takes **TOTP as a field** → fully headless daily login. Tokens die at **midnight**.
- Limits: order APIs **9/s cumulative**, 500/min, 1,000/hr; feed **1,000 token-subscriptions** per session (and a token counts once *per mode*); 3 connections.
- **Static IP mandatory at app registration** (max 5, changeable once a week).
- **Cost: API free**. Brokerage **₹20/order**.
- Measured: warm median 71.2 ms but best-case 22.9 ms — the **jitteriest** host in the test.

4.4 Shoonya (Finvasia)

- REST `api.shoonya.com`; one socket carries quotes **and** order updates (`t:"om"`), fills pushed with price/qty — **no subscribe step**.
- Session valid **for the trading day only, no refresh endpoint**; docs offer a **Selenium headless login** as the unattended path.
- **Static IP whitelisting mandatory** — 1 primary + 1 backup only.
- Same trap as XTS: **HTTP 200 for rejected orders** — you must check `stat == "Ok"`.
- Limits: ~10 orders/s; market data ~1/s per instrument; ceilings "may be tuned without notice".
- **Cost: API free**. Brokerage **₹5 per order** — the cheapest here.
- Measured: **fastest host in the test (23.7 ms)**.
- Doc conflict: its FAQ says `/NorenWClientTP/`, its docs say `/NorenWClientAPI/` — probe before building.

4.5 Alice Blue (ANT v2)

- REST `a3.aliceblueonline.com`; feed `wss://ws1.aliceblueonline.com/NorenWS`; **order-notify socket** plus optional **webhook**.
- Login is a **browser redirect** (`authCode` → SHA-256 → session). Token life **not stated**; treat as daily. No documented headless path.
- **Best-published rate limits of the field**: orders 10/s, 300/min, 3,000/hr, **200,000/day**; data 20/s; quotes 5/s.
- Contract master per exchange, refreshed **08:00 daily**; V1 endpoints retire **5 Oct 2026**.
- **Cost: API free for life**. Brokerage **₹20/order or 0.05%, whichever lower**.
- Feed needs a **50-second heartbeat**.

4.6 Kotak Neo

- **Caveat** — **Measured: the documented gateway host does not resolve** (`gw-napi.kotaksecurities.com` → NXDOMAIN), and `documentation.kotaksecurities.com` is dead too. The REST base is **assigned at login** (`baseUr1` in the response), so this is survivable — but the published entry points are stale.
- Login: **TOTP + MPIN, fully headless** — the cleanest unattended login of the field. Token life **not stated**; `403 Invalid session` is the expiry signal.
- Orders `POST /quick/order/rule/ms/place` → `nOrdNo`. Order/position socket `wss://<baseUr1>/realtime` with a **raw-string auth frame**.
- **Market orders are prohibited for retail algo trading (SEBI)**: they are auto-converted to **limit orders with 0.5–2 % protection**. Our engine sends market orders — this changes behaviour.
- Limits: 10 orders/s; per-endpoint REST quotas **not published**.
- Feed: 3,000 tokens total, 200 per request. The old feed was **deprecated 15 Sep 2026**, the v2 SDK **archived 10 Sep 2026** — anything older than that is stale.
- **Cost: API free; ₹0 brokerage on API orders** (statutory charges still apply) — the cheapest of all.
- Only broker here that publishes latency: **<50 ms** order execution, plus per-endpoint figures (login 367 ms, validate 134 ms, quotes 289 ms).

4.7 Jainam

- The link Ayush sent is **not XTS** — it is Jainam's own **ProTrade (Noren-based)** API: REST `protrade.jainam.in`, order feed `wss://protrade.jainam.in/omo/odrest/websocket` (**1-minute heartbeat**), market feed `wss://ws.jainam.in/NorenWSTP/`.
- Login is a **browser SSO redirect**; token life not stated.
- Limits: **order APIs "NOT LIMITED"**; everything else 1,800 requests / 15 minutes.
- Contract master per exchange, refreshed **08:00 daily**.
- Cost: API free.** Brokerage **₹20/order**.
- Jainam **also** sells Symphony XTS (dealer and retail), login window 08:30–15:28 — i.e. the same API family we already run.

4.8 Sharekhan (Mirae Asset Sharekhan)

- Measured: **slowest of every broker tested** — warm 101 ms, cold 403 ms, TLS 170 ms. Their marketing claims "up to 25 ms" per trade; our own network numbers are 4× that before any order is even sent.
- REST `api.sharekhan.com/skapi/services/...`; market feed `wss://stream.sharekhan.com/skstream/...`.
- Login is **browser + 2FA**, Kite-style (`request_token` → AES-decrypt with the secret → access token). **JWT valid 24 h, no refresh, no documented headless path** — so somebody logs in by hand every morning.
- Orders: `POST /skapi/services/orders`; modify and cancel are the **same POST path** with `requestType` NEW/MODIFY/CANCEL. **Whether the order id comes back synchronously is not published** — no sample response exists anywhere.
- Order updates ride the same socket under an `ack` key whose semantics are **undocumented**.
- Limits: "up to **30 orders/second**" (highest claimed here). Their own FAQ page for this is literally unfilled — "you will be able to place **XX** orders per second". Per-minute/day limits not stated.
- Instruments: scrip master is an **endpoint** (`GET /skapi/services/master/{exchange}`), not a file; size and refresh cadence not stated.
- Cost: API free. Options brokerage ₹39 per lot per side** — nearly as bad as Lakshmishree's ₹40, and per *lot*, so a 13-lot round trip is ~₹1,014. (A ₹20/trade F&O plan is reported to exist; plan-dependent, confirm on the account.)
- Wart worth naming: Sharekhan's **own** Python SDK sets `ssl.CERT_NONE` on the websocket — TLS certificate verification switched off — and has no retry/backoff.
- Documentation is a JavaScript single-page app that returns an empty shell to any fetch; the SDK source on their GitHub org is effectively the real documentation.

4.9 Groww

- Measured: warm 45.1 ms, behind Cloudflare. Their blog claims an "average response speed of 65 ms"; no SLA.
- REST `api.groww.in/v1`; instrument CSV at a fixed URL; the live feed is **NATS over WebSocket** whose **host is not published** — it is buried in their SDK.
- Three login paths: a token pasted from the profile page (**expires daily at 06:00**, useless unattended), API key + secret with `SHA256(secret+timestamp)`, and a **TOTP** path the SDK page labels "**No Expiry**". **Caveat — Their docs contradict each other** — the curl page says both key flows still need "daily approval on the Groww Cloud API Keys page". Must be probed live before trusting it.
- Orders: `POST /v1/order/create` → **returns groww_order_id synchronously**, plus modify/cancel/status/trades. Order and position updates push over the NATS feed.
- Limits (published clearly, unlike most): auth 5/s 30/min · **orders 10/s, 250/min** · live data 10/s 300/min · non-trading 20/s 500/min · token endpoint **150 calls per 24 h**. Feed "up to 1,000 instruments" (their blog says 3,000/user — another internal contradiction); REST quote batching max 50.
- Instruments: 19-column CSV including `exchange_token`, `lot_size`, `expiry_date`, `strike_price`, `freeze_quantity` — the cleanest master of the field for our purposes.
- Cost: free tier covers all APIs except live data; ₹499 + tax/month if you want their data** (we don't — we run TrueData). Brokerage **₹20 per executed order, flat** — best of these three for multi-lot.
- Restriction that matters: CASH + F&O only, no MCX**. Not an issue for NIFTY/SENSEX.
- SDK is PyPI-only (`growwapi`); there is **no official public GitHub repo**, and the npm package of that name is third-party.

4.10 Choice FinX

- Measured: **second fastest (24.5 ms)**, AWS Mumbai, answered 200 on an unauthenticated probe. Their marketing is the most aggressive of anyone: **7.4 ms round trip, p50 7 ms, p99 9.5 ms, "<10 ms" SLA**, co-located at NSE/BSE/MCX. Unaudited, and our own 24.5 ms is the only figure either of us can verify.
- Caveat — Choice publishes no public developer documentation at all** — `finx.choiceindia.com/api-portal` 404s and `docs.finx.choiceindia.com` does not resolve. Everything below comes from a Choice-issued PDF walkthrough, a published SDK and one third-party integration.
- Hosts `finxomme.choiceindia.com` / `finx.choiceindia.com` — the "omne" name betrays an **Omnesys/NEST lineage. It is not XTS**: proprietary `api/OpenAPIV1/...` namespaces, `VendorId` + `Bearer` + `SessionId` auth, and market data delivered as **FIX 3.0 over raw TCP with Zlib**, host and port handed out at login — not socket.io. Treat any "Choice = XTS" claim as wrong.
- Unattended login: yes, and it is the cleanest of the three** — three POSTs (`LoginTOTP` → `GetClientLoginTOTP` → `ValidateTOTP`), and **Choice returns the OTP through the API itself** rather than pushing it to a phone. API key is a JWT you generate in the dashboard with a **1-day or 1-month validity**; the session is day-scoped.

- **Cost: API free forever, no subscription, no per-call fee.** Brokerage ₹25 per lot on NSE equity options — per lot again, so a 13-lot round trip is ~₹650.
- **Three restrictions that would change our engine:** 1. **Market orders are not supported** — limit and stop-loss-limit only. 2. **Prices must be sent in paisa** (₹ × 100). 3. **F&O quantities are in total shares, not lots.**
- **Static-IP binding is mandatory** — one declared IP per key; VPNs, proxies and serverless egress are rejected, and the key must be rotated when the IP changes.
- Unknowns that block any estimate: the **place-order path itself is not published**, nor the order-websocket URL, nor subscription caps, nor any rate limit (they do send `Retry-After`, so throttling exists). Broker login window reported as **08:30–15:28 IST**.

4.11 "Sahi Pro" — not a broker

- The "Documentation APIs" link is **Sahi Pro by Tyto Software**, a **software test-automation tool**. Its "Documentation APIs" are `_startDocumentation()` / `_stopDocumentation()` for generating test reports. Nothing to do with trading.
- The **broker** named *Sahi* (sahi.com) is a different company (₹10/order F&O). Its **API is not public yet**.
- **Conclusion: nothing to integrate here today.**

4.12 Fyers v3 (compared in full on 18 Sep — see that report)

- REST `api-t1.fyers.in`; order socket `wss://socket.fyers.in/trade/v3`.
- Token expires **daily**; OAuth auth-code flow.
- Limits: 10/s, 200/min, 100,000/day — and **3 per-minute breaches = blocked for the day**.
- **Cost: API free.** Brokerage ₹20/order.
- Symbol master 84 MB daily; its `exToken` equals the exchange token we already map.

5. One-line scorecard

Broker	Measured warm	Brokerage (options)	API fee	Unattended daily login	Order-fill push	Static IP	Fit for us
Lakshmishree (ours)	29.5 ms	₹40/lot ≈ ₹1,040 per 13-lot round trip	free	token-based, works	socket.io (unused)	no	Working, but by far the most expensive
Kotak Neo	host unresolved	₹0	free	Yes (TOTP+MPIN)	yes	portal-managed	Cheapest, but no market orders for algo
Shoonya	23.7 ms	₹5/order	free	Selenium only	yes (auto)	mandatory, 2 IPs	Cheapest usable, fastest — IP rule is the catch
Fyers	42.3 ms	₹20/order	free	daily OAuth	yes	no	Good all-rounder
Dhan	52.9 ms	₹20/order	free + ₹499/mo data	Yes (TOTP)	dedicated socket	mandatory	Good, if the static IP is fine
Angel One	71.2 ms (jittery)	₹20/order	free	Yes (TOTP)	yes	mandatory	Fine; feed limits tight
Alice Blue	29.6 ms	₹20/order or 0.05%	free	browser login	yes + webhook	not stated	Fast, best-published limits
Zerodha Kite	36.3 ms	₹20/order	₹500/mo	No — manual daily	yes (same socket)	no	Ruled out by the manual login
Jainam ProTrade	not probed	₹20/order	free	browser SSO	yes	not stated	Order APIs unlimited; SSO is the catch
Groww	45.1 ms	₹20/order flat	free (₹499/mo only for data)	TOTP — docs contradict	yes (NATS)	not stated	Cheapest flat rate; no MCX; verify the login
Choice FinX	24.5 ms	₹25/lot ≈ ₹650 per round trip	free	Yes (API returns its own OTP)	yes	mandatory	Best login story, but no market orders
Sharekhan	101.2 ms	₹39/lot ≈ ₹1,014	free	No — browser + 2FA	undocumented ack	not stated	Slowest, expensive, docs unreadable
Sahi Pro	—	—	—	—	—	—	Not a broker

Cost, made concrete. Same 13-lot NIFTY round trip we actually traded on 15 Sep:

Broker	Structure	Cost of that round trip
Kotak Neo	₹0 on API orders	₹0 (+ statutory)
Shoonya	₹5/order	₹10
Groww · Fyers · Kite · Dhan · Angel · Alice Blue · Jainam	₹20/order	₹40
Choice FinX	₹25/lot	₹650
Sharekhan	₹39/lot/side	₹1,014
Lakshmishree (ours today)	₹40/lot/side	₹1,040

That single column is the whole argument. On the trade we booked, brokerage alone (₹1,040 of the ₹1,397.95 total charge) turned a small loss into a ₹1,398 loss.

5b. Documentation quality — because this decides how long an integration takes

Ayush asked about "actual effectiveness" versus the documentation. Ranked by how much of what we need is actually written down:

Broker	Docs	What that costs us
Zerodha Kite · DhanHQ · Angel · Groww · Fyers	Complete, public, readable	estimate is reliable
Alice Blue · Shoonya	Good, but Shoonya's own FAQ and docs disagree on the URL path	one probe to settle
Symphony XTS (Lakshmishree)	Complete on shape, silent on error behaviour and on charges	we discovered HTTP-200 errors in production
Jainam ProTrade	Adequate; token lifetime not stated	small unknown
Kotak Neo	Published hosts are dead (gw-napi... NXDOMAIN, docs site gone); SDK archived 10 Sep 2026	can't even start without support contact
Sharekhan	JavaScript shell — returns nothing to a fetch. Their SDK source is the real doc. Rate-limit FAQ literally says "XX orders per second"	order-id behaviour is an unknown
Choice FinX	None public at all. The place-order path is not published anywhere	cannot be estimated without their sales team

A broker whose documentation we cannot read is not a cheaper broker — it is an integration priced at "unknown".

6. What the documentation never tells you (lessons from our own live test)

1. **Brokerage is invisible to every API.** Only the contract note shows it. We found ours was **₹40/lot**, not the ₹17/order we had modelled — a 30× error that no API call would have revealed.
2. **HTTP 200 can mean failure.** True for Lakshmishree (dead token) and Shoonya (rejected order). Always branch on the body.
3. **Broker-side P&L fields can be dead.** Lakshmishree's MTM stayed 0.00 and margin lagged minutes. Compute your own.
4. **Published hostnames rot.** Kotak's documented gateway does not resolve; its feed and SDK were replaced in September 2026.
5. **The login is the fragile part,** not the order. Daily tokens, TOTP, static IPs and browser redirects decide whether a system can run unattended at 09:15.
6. **Since 1 April 2026 every API order must carry a broker-issued Algo ID** (SEBI). Below 10 orders/second a generic exchange ID applies — we are far below it, but the ID still has to be registered with whichever broker we use.

7. Recommendation

1. **Fix the cost first, not the technology.** Ask Lakshmishree to move from ₹40/lot to a ₹20-per-order plan. That one conversation is worth ₹1,000 per round trip — more than any integration on this list, and it costs nothing to have.
2. **If they refuse, the shortlist is three, in this order:** * **Fyers (₹20/order)** — free API, readable docs, order push socket, no static-IP requirement, and the integration is already scoped at **44–60 hours**. Lowest-risk move. * **Shoonya (₹5/order)** — cheapest usable and the fastest host we measured (23.7 ms), *if* we can guarantee a fixed egress IP (they allow exactly 2) and accept a Selenium login. * **Groww (₹20/order)** — the best public documentation of the field and a clean instrument master; gated on one question we can settle with a single probe: does the TOTP token really survive without a daily dashboard click?
3. **Talk to, don't build for, Kotak Neo.** ₹0 brokerage is genuinely the best number here, but market orders are prohibited for retail algo accounts (auto-converted to protected limit orders — that changes how our engine fills), and their published gateway host does not exist in

DNS. Confirm both before anyone writes code.

4. **Rule out for now:** * **Zerodha Kite** — the daily login is manual by policy; we cannot run unattended at 09:15. * **Sharekhan** — slowest measured by 3×, ₹39 *per lot*, browser-only login, and documentation we cannot read. * **Choice FinX** — fast and free, but **no market orders**, prices in paisa, quantities in shares, mandatory static IP, ₹25/lot, and no published place-order endpoint. Too many engine changes for a worse rate than Fyers. * **Sahi Pro** — not a broker.
5. **Whatever we pick, re-measure with a funded account.** Nothing in this report measures order matching at the exchange; only a real burn-in like 15 September shows the truth, and that burn-in is exactly what taught us the ₹40/lot number that no API would have revealed.

8. Sources

Broker documentation: Kite Connect v3 · DhanHQ v2 · Angel One SmartAPI · Shoonya API docs · Alice Blue v2 API · Kotak Neo (SDK repo + kotakneo.com) · Jainam ProTrade apidocs · Symphony XTS v2 Interactive/Market Data · Fyers API v3 · Sharekhan · Groww · Choice FinX. Charges pages of each broker.

Our own evidence: docs/burn-in-2026-09-15/LIVE-BROKER-BURN-IN-2026-09-15.md · contract note 322782 (15 Sep 2026) · docs/BROKER-COMPARISON-FYERS-vs-LAKSHMISHREE-2026-09-18.md · backend/app/algo/broker/xts_interactive.py · latency probes run 19 Sep 2026.